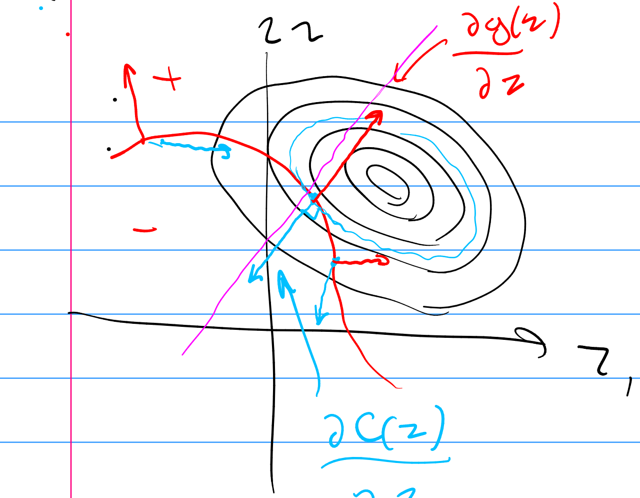
$C(z)$ $\quad z \in$ curve $\quad$ curve $\qquad g(z) = 0$



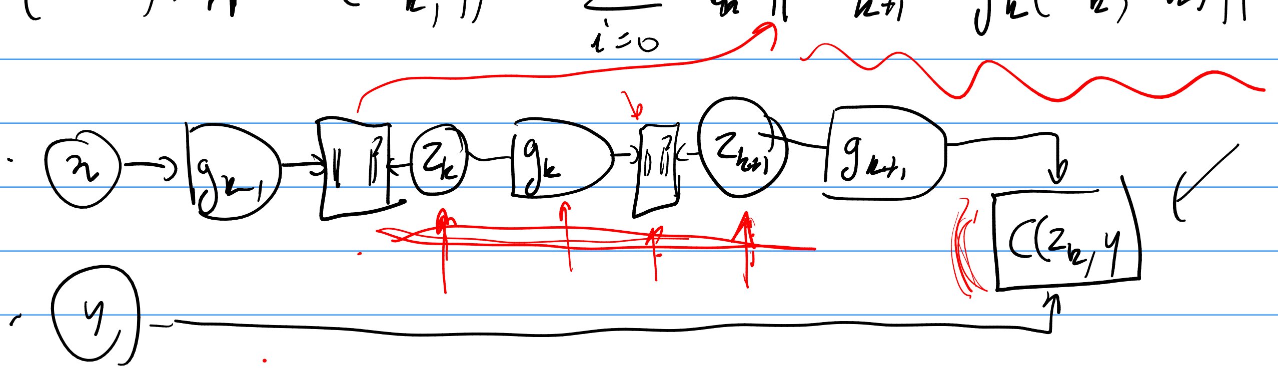
$$L(z, \lambda) = C(z) + \lambda^T g(z)$$

$$\forall \lambda \quad \frac{\partial C(z^*)}{\partial z} \propto \frac{\partial g(z)}{\partial z}$$

$$L'(z, \lambda) = C(z) + \alpha \| g(z) \|^2$$

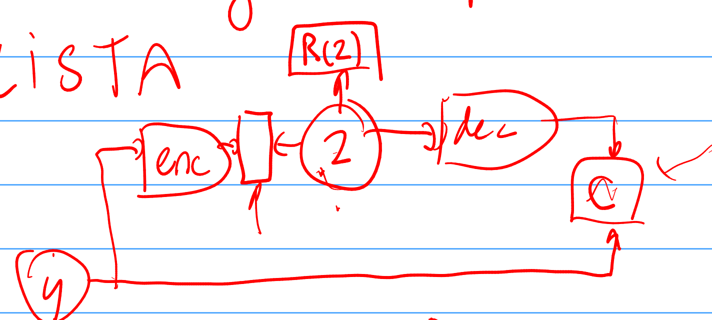
$$L(x, y, z, \lambda, w) = C(z_k, y) + \sum_{i=0}^{k-1} \lambda_k^T \left(z_{k+1} - g_k(z_k, w_k) \right)$$

$$L'(\quad) = \lambda'^T C(z_k, y) + \sum_{i=0}^{k-1} \| z_{k+1} - g_k(z_k, z_k) \|^2$$



Target Prop

LISTA



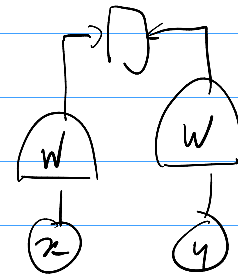
amortized inference

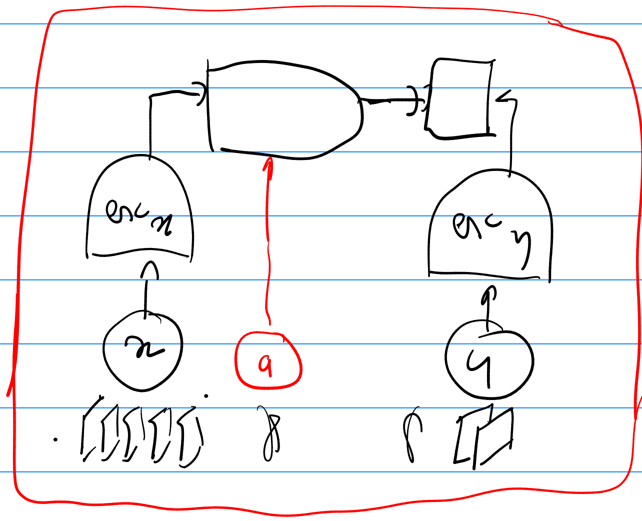
SSL ⟷ "unified architectures"

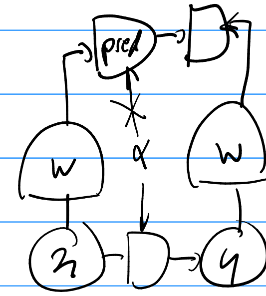
SSL : non-contrastive method VICReg BT joint embedding

SSL → learning representations

SSL → learning forward models

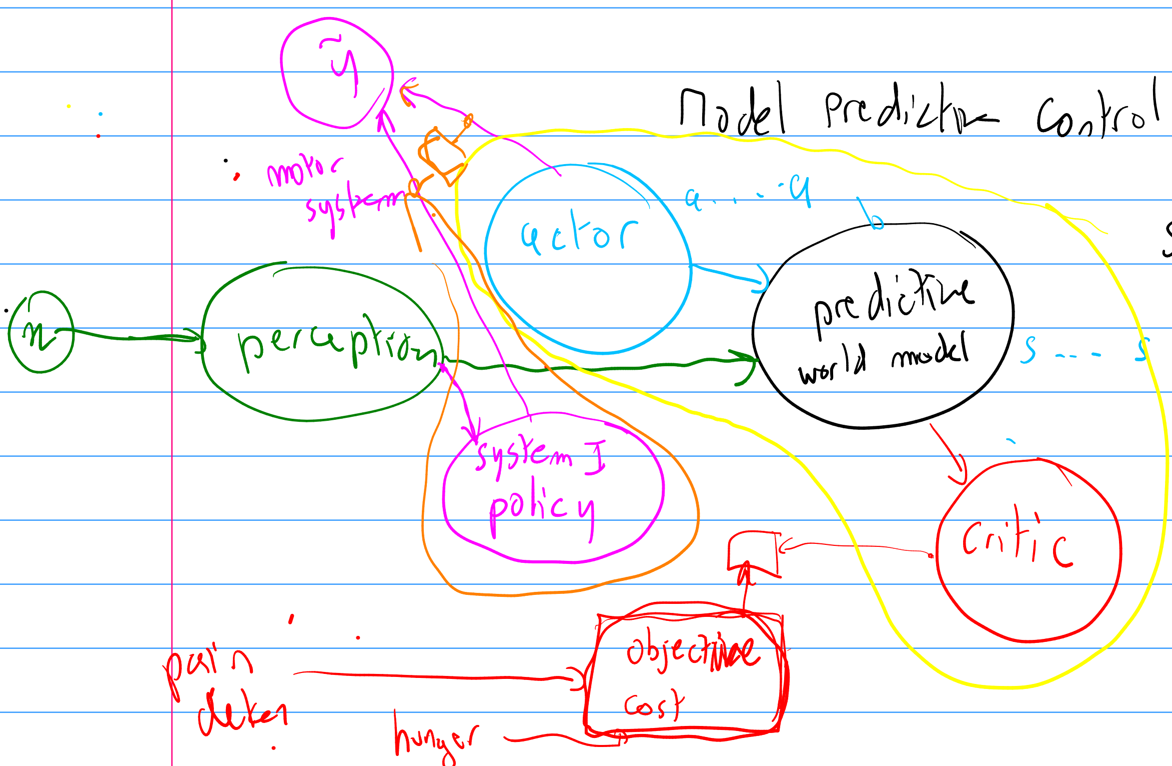






Model predictive control

Daniel Kahneman



System II

planning

conscious action

⋮